INCH-POUND

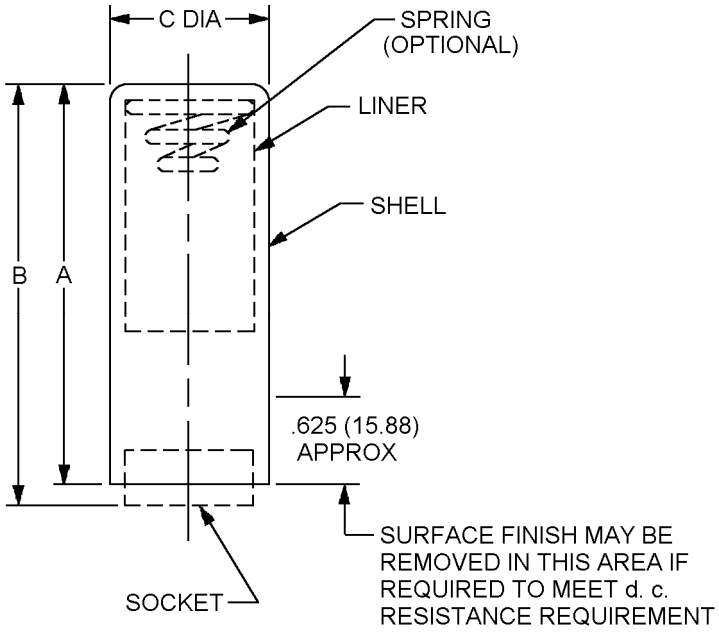
MIL-DTL-24251/6D
15 February 2008
SUPERSEDING
MIL-DTL-24251/6C
14 May 2003

DETAIL SPECIFICATION SHEET

SHIELD, ELECTRON TUBE, HEAT DISSIPATING,
7 AND 9 PIN, MINIATURE (RETROFIT TYPE ONLY)

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24251.

This specification is approved for use by all Departments and Agencies of the Department of Defense.




NOTES:
1. All dimensions are in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest 0.01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. Unless otherwise specified, tolerance is 2 places ± .010 (0.25 mm); 3 places ± .005 (0.13 mm).
5. MIL-DTL-24251/5 (shield) was formerly MS24233 (USAF), 25 Apr 1963.

FIGURE 1. Dimensions and configuration.

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TABLE 1. Characteristics.

| Dash no. | Heat dissipation temp. reduction in °C (minimum) | A -.062 (1.57) | B Max | C dia ±.015 (0.38) | Weight grams max | Tube envelope | | Tube socket ref 2/ |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Style | Length nom | |
| 1 | 25 | 1.500 (38.10) | 1.625 (41.28) | .875 (22.23) | 11.0 | T-5 1/2 | 1.125 (28.58) | MIL-DTL-12883/2 |
| 2 | 40 | 2.000 (50.80) | 2.125 (53.98) | .875 (22.23) | 15.0 | T-5 1/2 | 1.500 (38.10) | |
| 3 | 45 | 2.500 (63.50) | 2.625 (66.68) | .875 (22.23) | 19.0 | T-5 1/2 | 2.000 (50.80) | |
| 4 | 25 | 1.500 (38.10) | 1.625 (41.28) | 1.000 (25.40) | 12.0 | T-6 1/2 | 1.125 (28.58) | MIL-DTL-12883/3 |
| 5 | 55 | 2.000 (50.80) | 2.125 (53.98) | 1.000 (25.40) | 16.0 | T-6 1/2 | 1.562 (39.67) | |
| 6 | 45 | 2.500 (63.50) | 2.625 (66.68) | 1.000 (25.40) | 20.0 | T-6 1/2 | 2.000 (50.80) | |
| 7 1/ | 45 | 2.750 (69.85) | 2.875 (73.03) | 1.000 (25.40) | 22.0 | T-6 1/2 | 2.812 (71.42) | |

1/ This shield is for use with noval tubes which are longer than usual tubes.

2/ Not recommended for new design.

REQUIREMENTS:

The shield, locking mechanism, liner, and other parts shall be designed to meet the requirements in MIL-DTL-24251, and to fit the tube envelope and tube socket specified in table I.

Dimensions and configuration: See figure 1 and table I. No special retainer (base) or adapter will be permitted.

Material:

Spring (when used): Beryllium copper, ASTM-B194, ASTM-B196, and ASTM-B197.
Shell: Aluminum alloy, SAE-AMS-QQ-A-250 (aerospace and unspecified applications), SAE-AMS-QQ-A-250/2, or ASTM-B209 (non-aerospace applications).
Liner: Beryllium copper, ASTM-B194.

Finish:

Spring (when used): Black cadmium plate, SAE-AMS-QQ-P-416, class 1, type II.
Shell: Black anodize, type II in accordance with MIL-A-8625, black matte in accordance with MIL-DTL-5541, optional finish: Henderlube or equal.
Liner: Black cadmium plate, SAE-AMS-QQ-P-416, class 3, type II, optional finish: Henderlube or equal.

Construction: The spring shall be held in place by friction or by mechanical means if the maximum diameter is smaller than the inside diameter of the shield. The tension shall be such as will insure maximum compression of .125 inch (3.18 mm) over seated tube. The force required to compress spring to .375 inch (9.53 mm) shall be within 1.5 to 2.5 pounds. Spirals shall not touch each other. The end of the coil spring shall not contact the electron tube envelope.

Installation force shall not exceed 20 pounds.

Engaging torque: 20 inch-pounds maximum.



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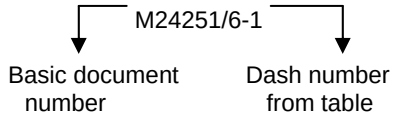
The Part or Identifying Number (PIN) consists of the basic number of this specification sheet, preceded by the letter "M", and a dash number taken from table I.

Example:




Referenced Documents: In addition to MIL-DTL-24251, this document references the following:
MIL-DTL-5541
MIL-A-8625
MIL-DTL-12883/2
MIL-DTL-12883/3
ASTM-B194
ASTM-B196/B196M
ASTM-B197/B197M
ASTM-B209
SAE-AMS-QQ-A-250
SAE-AMS-QQ-A-250/2
SAE-AMS-QQ-P-416

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2008-010)

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.



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